

Pradyumn Tendulkar

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AI Engineer with production experience building LLM applications, multi-agent pipelines, and RAG systems using Claude, OpenAI, and AWS. MS in Data Science (4.0 GPA). 2 peer-reviewed publications. Open to relocation.

SKILLS

LLM & GenAI: RAG, Multi-Agent Systems, Agentic AI, LangChain, LangGraph, LlamaIndex, Vector Databases, Claude API, OpenAI API, Gemini, Fine-Tuning, Prompt Engineering, LLM Evaluation, Guardrails, Model Context Protocol (MCP)

ML & NLP: PyTorch, Hugging Face, Transformers, TensorFlow, scikit-learn, spaCy, NLTK

Cloud & MLOps: AWS (Bedrock, SageMaker, Lambda, Step Functions, Textract, Comprehend, CloudFormation), GCP (Vertex AI, Cloud Run), Docker, Kubernetes, CI/CD, MLOps, FastAPI

Languages & Tools: Python, TypeScript, SQL, Pydantic, Pandas, NumPy, PostgreSQL, React, Next.js, Git

EXPERIENCE

AI Engineer, CrossingLegal, Boston, MA Aug 2025 – Dec 2025

- Architected multi-agent validation pipeline using **AWS Step Functions**, **Lambda**, and **Bedrock Claude API** for vision-based PDF-to-JSON extraction, reducing H-1B processing from 5 days to 10 minutes with production guardrails.
- Constructed RAG system with Titan embeddings, in-memory vector database, and cosine similarity retrieval to resolve ambiguities across 50+ Form I-129 fields, achieving 99% extraction accuracy.
- Integrated LLM validation layer using **OpenAI API**, Pydantic schema enforcement, and **Textract**-based knowledge base, cutting manual verification effort by 90%.

AI Engineer Intern, Boredm LLC, Phoenix, AZ Jun 2025 – Aug 2025

- Fine-tuned **GPT-4o-mini** via **OpenAI API** on 10,000+ balanced examples for 58-entity NER extraction, implementing Pydantic schema validation to achieve 99.1% production accuracy on soil lithology descriptions.
- Engineered synthetic data pipeline using **Python-TypeScript** hybrid architecture with dependency-aware generation and structural variants, reducing manual annotation time by 90%.

Research Assistant, Worcester Polytechnic Institute, Worcester, MA Oct 2024 – May 2025

- Developed zero-shot summarization framework using **GPT-4**, **Gensim**, **GloVe embeddings**, and **NLTK** preprocessing to generate abstractive summaries from 5,000+ Amazon reviews, published at **AMCIS 2025**.
- Attained 0.94 cosine similarity with Amazon's proprietary summaries using **scikit-learn** cosine similarity, iterative prompt refinement, and **BERTScore** validation on **Playwright**-scraped review data.

AI Engineer Intern, SmartStream Technologies, India May 2023 – Jul 2023

- Deployed end-to-end Named Entity Recognition pipeline using **AWS Textract**, **SageMaker Ground Truth**, and **Comprehend** on PDF/Word documents, achieving 99% accuracy.
- Replaced legacy rule-based system with **Comprehend** ML workflows and **CloudFormation**-provisioned annotation infrastructure, reducing entity extraction cycles from 4 months to 3 weeks (80% improvement).

PROJECTS

RAG-Powered Legal Research Engine [Link](#)

- Created full-stack RAG application using **LangGraph** (6-node agentic pipeline), **FastAPI**, **GPT-4o**, and **Pinecone** vector database for legal case research, with dual-layer confidence scoring and **Mistral** fallback, hosted on **Vercel/Railway**.

AI Content Pipeline with Self-Improving Hooks [Link](#)

- Designed multi-agent pipeline using **Claude**, **Gemini**, and **Tavily** APIs with 3-phase chain (research, draft, refine) and self-improving hook analysis learning from engagement metrics, aggregating 14 news sources.

Generalizable Vision-to-JSON Extraction Framework [Link](#)

- Built vision-based PDF extraction using **OpenAI Vision API** (GPT-4.1) with adaptive DPI rendering, supporting 12 document types (I-129, passports, resumes) with JSON schema enforcement, served on **Hugging Face Spaces**.

Insurance Policy Analysis Pipeline [Link](#)

- Orchestrated 3-stage insurance policy analysis pipeline using **Next.js 16** and **Claude Sonnet 4** vision for structured PDF extraction, parallel coverage gap analysis and talking point generation across 3 edge API routes.

PUBLICATIONS

"Predictive Modeling of Water Turbidity Using Remote Sensing Indices" – IEEE 2026 (Accepted)

"Zero-Shot Abstractive Summarization of Amazon Product Reviews Using GPT-4" – AMCIS 2025 [Link](#)

EDUCATION

Worcester Polytechnic Institute, Worcester, MA - MS in Data Science (GPA: 4.0) Aug 2024 – Dec 2025

- Courses:* Statistical Methods for Data Science, Deep Learning, Natural Language Processing, MLOps.

Vellore Institute of Technology, India - BTech in Computer Science and Engineering (GPA: 3.56) Aug 2021 – May 2024